1. OVERVIEW TAB

What This Tab Represents

Think of this page as:

```
Cluster Identity Card
```

Everything here describes the Kubernetes Control Plane.

What to Observe in Screenshot

Cluster Status

```
Active
```

Meaning:

```
API Server Running
Scheduler Running
Controller Manager Running
etcd Running
```

Kubernetes Version

```
1.35
```

This is NOT:

```
EC2 Version
Linux Version
Docker Version
```

This is:

```
Control Plane Version
```

API Server Endpoint

Example:

https://xxxxx.eks.amazonaws.com

This endpoint becomes the target of:

```
kubectl get nodes
kubectl get pods
kubectl apply -f deployment.yaml
```

OIDC Provider

Example:

https://oidc.eks.ap-south-1.amazonaws.com/id/xxxx

Future Use:

```
Pod Identity
IRSA
IAM Roles for Pods
```

Learning Note

At this stage:

```
Control Plane Exists
Worker Nodes Do Not Exist
```

Interview Note

Question:

```
What components exist immediately after EKS cluster creation?
```

Answer:

```
API Server
Scheduler
Controller Manager
etcd

NOT Worker Nodes
```

2. RESOURCES TAB

What This Tab Represents

This page displays Kubernetes objects stored inside:

```
etcd
```

What to Observe

CoreDNS Pods

Example:

```
coredns-xxxxx
Pending

coredns-yyyyy
Pending
```

Why Are They Pending?

Current architecture:

```
CoreDNS Pod
        |
Needs Node
        |
Nodes = 0
        |
Pending
```

Learning Note

This is the first proof that:

```
Pods cannot run without Nodes
```

Architecture

```
Deployment
     |
ReplicaSet
     |
Pods
```

CoreDNS already exists as a Deployment.

3. COMPUTE TAB

What This Tab Represents

Where Kubernetes gets compute power.

Current Observation

```
Nodes = 0
Node Groups = 0
Fargate Profiles = 0
```

Relationship Explained

This answers:

Where does Auto Scaling Group fit?

Architecture:

```
Managed Node Group
        │
        ▼
Auto Scaling Group
        │
        ▼
EC2 Instances
        │
        ▼
Kubernetes Nodes
        │
        ▼
      Pods
```

Learning Note

AWS hides the ASG from most Kubernetes users.

But it still exists.

Interview Note

Question:

```
Does EKS automatically create EC2 instances?
```

Answer:

No

```
Cluster Creation
≠
Node Creation
```

4. NETWORKING TAB

What to Observe

VPC

```
vpc-xxxx
```

All cluster resources live here.

Service CIDR

```
10.100.0.0/16
```

Used by:

```
Kubernetes Services
```

NOT:

```
EC2 Instances
```

Subnets

```
ap-south-1a
ap-south-1b
ap-south-1c
```

Provides Multi-AZ resiliency.

Cluster Security Group

Created automatically by EKS.

Used for:

```
Control Plane ↔ Nodes
```

communication.

Learning Note

Distinguish:

```
VPC CIDR
```

from:

```
Service CIDR
```

Very common confusion.

5. ADD-ONS TAB

CoreDNS

Provides:

```
DNS Resolution
```

Example:

```
backend-service
       |
       ▼
10.100.x.x
```

kube-proxy

Provides:

```
Service Routing
```

Amazon VPC CNI

Provides:

```
Pod Networking
```

and Pod IP allocation.

Golden Rule

Remember:

```
CoreDNS
      ↓
Name Resolution

kube-proxy
      ↓
Traffic Routing

VPC CNI
      ↓
Pod Networking
```

6. CAPABILITIES TAB

Argo CD

GitOps platform.

```
GitHub
     │
     ▼
Argo CD
     │
     ▼
Kubernetes
```

ACK

AWS resources through Kubernetes.

Example:

```
kubectl apply
      │
      ▼
Creates S3 Bucket
```

KRO

Higher-level Kubernetes orchestration.

Learning Note

Skip these until Kubernetes fundamentals are mastered.

7. ACCESS TAB

Authentication Mode

```
EKS API + ConfigMap
```

Supports:

```
Modern Access Entries
Legacy aws-auth
```

IAM Access Entries

Controls:

```
Who can access cluster?
```

Pod Identity

Controls:

```
What AWS resources can Pods access?
```

Architecture

```
Pod
 |
 ▼
IAM Role
 |
 ▼
S3
Secrets Manager
DynamoDB
```

8. OBSERVABILITY TAB

Control Plane Logs Enabled

```
API Server
Audit
Authenticator
```

Purpose

Answer:

```
What happened?
Who did it?
When did it happen?
```

Future Tools

```
Prometheus
Grafana
CloudWatch
```

9. UPDATE HISTORY & BACKUPS

Update History

Tracks:

```
Cluster Upgrades
Add-on Upgrades
Configuration Changes
```

Backups

Protects:

```
Deployments
Services
Secrets
ConfigMaps
```